

K AZAAN AHAMED

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SUMMARY

Final-year student with a problem-driven mindset and hands-on exposure across VLSI Physical Design, Embedded Systems and Machine Learning. Through academic and personal projects, developed practical experience in applying engineering concepts to real-world problems while strengthening technical and analytical abilities. Alongside academics, leads two student clubs and manages clients for content creation. Brings strong leadership, analytical thinking, communication and teamwork skills, with the adaptability to contribute effectively across both VLSI and embedded engineering environments.

SKILLS SUMMARY

Hardware: Arduino Uno, ESP32, Sensors & Modules, Circuit Design

Simulation Tools: Cadence (Virtuoso, Innovus, Genus), MATLAB, ModelSim, NetSim, OptiSystem

Languages: Verilog HDL, C Programming, Linux (Basic Commands)

Multimedia Tools & Skills: Photography, Content Creation, Canva, CapCut, AI Agents, Excel

Soft Skills: Leadership, Problem-Driven Analysis, Communication, Adaptability, Teamwork & Management

EDUCATION

Vellore Institute of Technology

Bachelor of Technology – Electronics and Communication Engineering | **CGPA: 8.42**

Vellore, India

2023 – 2027

Narayana Group of Schools & Colleges

Intermediate (MPC) | **Score: 94%**

2020 – 2022

SSC (Class 10th) | **Score: 94%**

2018 – 2020

CERTIFICATIONS & EXPERIENCE

VLSI Physical Design Intern – Maven Silicon Pvt. Ltd.

May 2026 – Jul 2026 | Remote

- Completed certification and gained practical exposure to ASIC design and the physical implementation flow.
- Worked on a final project implementing an SPI module using Qflow.
- Gained hands-on experience in taking a digital design from RTL to a final chip layout through a complete design flow.

PROJECTS

ML-Based Routing & Congestion Predictor for RV32I Processor

Aug 2026 – Present (Ongoing)

- Developing an ML model to predict routing congestion in an RV32I processor during the RTL-to-GDSII flow.
- Congestion is a major cause of delays, and Cadence tools only flag that it exists without suggesting how to resolve it, so the model predicts congestion early and recommends the parameter changes needed to fix it.

Tech: Cadence (Virtuoso, Innovus, Genus), OpenROAD (Dataset), Antigravity, Google Colab, Gradient Boosting (ML Training)

Physical Design of Serial Peripheral Interface (SPI) using Qflow

May 2026 – Jul 2026

- Implemented the complete RTL-to-GDSII physical design flow for an SPI module using the Qflow EDA toolchain.
- Performed logical synthesis, placement, routing, timing analysis, DRC, LVS and final GDSII generation, and verified timing and physical design constraints.

Tech: Linux, Yosys, OpenSTA, Verilog HDL

Smart Rover: Multi-Function Robotic Car

Jan 2026 – Apr 2026

- Built an Arduino Uno-based robotic car using an HC-SR04 ultrasonic sensor for obstacle detection, an L298N motor driver for drive control, and an HC-05/06 module for Bluetooth communication.
- The car supports four operating modes: autonomous obstacle avoidance, Bluetooth control, serial terminal control and voice control.

Tech: Arduino Uno, C, Embedded Systems

Intelligent IV Therapy Monitoring and Automatic Flow Control System

Jan 2026 – Apr 2026

- Built an Intelligent IV Therapy Monitoring System that tracks IV fluid level and flow rate in real time, reducing the risk of an IV running empty or over-infusing.
- Nurses with heavy workloads often struggle to monitor multiple IV drips manually, so the system automates this and alerts staff when the fluid level is low.

Tech: Arduino Uno, ESP8266, C, LCD

ACHIEVEMENTS & EXTRACURRICULAR

Photography Head, SAHITI Club

- Led photography operations for **12+ university events**, coordinating a team of volunteers and ensuring consistent visual coverage for events with **1,200+ participants**.
- Managed event requirements, team allocation and on-ground execution to deliver timely and organized content.

Outreach Head, VRIKSH Club

- Led community outreach initiatives involving **50+ participants**, coordinating volunteers, planning activities and managing on-ground execution to ensure smooth and effective engagement.